

Maarij Alam

BScEng Electrical & Computer Engineering (Expected Nov 2026) · University of Cape Town

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PROFILE

Final-year ECE student with a strong foundation in software engineering, algorithms, and systems programming. I am comfortable working across the stack, from low-level embedded C through to cloud-deployed serverless backends, and I am drawn to problems where performance, correctness, and real-world behaviour under load actually matter. I enjoy working in teams where the code is taken seriously and there is something to learn from the people around me.

EDUCATION

BScEng Electrical & Computer Engineering *University of Cape Town, Cape Town* 2023 – Present

- Relevant coursework: Data Structures & Algorithms, OOP (Java), Embedded Systems (C), Computer Architecture, Linear Algebra, Control Systems, Signal Processing
- Elected Class Representative 2023–2024 and 2026; Orientation Leader (2026)

TECHNICAL SKILLS

Languages: Java, Python, C/C++, MATLAB, Bash

CS Fundamentals: Data Structures, Algorithms, SQL, OOP, Multithreading, Systems Programming, Complexity Analysis

Cloud & Backend: AWS (CDK, Lambda, API Gateway, Bedrock, DynamoDB), Node.js, Serverless, REST APIs

Tools & OS: Git/GitHub, Linux (Debian / Raspberry Pi), VS Code

SELECTED PROJECTS

Healthcare Voice-Triage PWA – AWS CDK, Node.js, Amazon Bedrock – Standard Bank UniHack 2026

- Architected the serverless backend for a multilingual voice-triage application using AWS CDK, integrating API Gateway, Lambda, and Bedrock for real-time LLM processing.
- Designed a no-PII DynamoDB data model and optimized Lambda invocations to run the full simulated patient load at under \$2 in total compute cost.

ECG Digitisation Mobile App (Currently Ongoing) – Python, ML, Mobile Development

- Designing a machine learning inference pipeline for clinical-grade ECG signal digitization targeting consumer mobile hardware, adhering strictly to on-device memory and latency constraints.

AVL Tree Analyser – Java, Data Structures

- Implemented a self-balancing AVL tree from scratch, benchmarking $O(\log n)$ insertion and deletion across 50,000+ entries to validate empirical worst-case complexity analysis.

Antarctic Remote Sensing Subsystem – C, Embedded Systems

- Developed reliable firmware and data acquisition logic for a remote sensing subsystem operating in extreme Antarctic conditions, emphasizing low-power scheduling for long-duration capture.

EXPERIENCE

Course Tutor – EEE3088F Engineering Design *University of Cape Town* Feb 2026 – June 2026

- Improved student design outcomes across a 100+ person engineering cohort by delivering structured technical feedback on design requirements, prototyping, and iterative system testing.

Mechatronics Systems Intern *MechatronicSystems.Group, Cape Town* Dec 2025 – Feb 2026

- Owned the end-to-end software development lifecycle (SDLC) to architect, build, and deploy a real-time drone state estimation application for operator monitoring.
- Developed a custom MATLAB GUI that integrated live OpenMV camera feeds with an Extended Kalman Filter (EKF) for real-time 3D position and velocity visualization.

Computer Vision Engineering Intern *AQUA Project, University of Cape Town* Jun 2025 – Aug 2025

- Architected a dual-camera failure-detection system utilizing multi-threaded blob-detection and edge-deployed AI models.
- Achieved continuous unattended defect detection by optimizing AI inference throughput on constrained edge hardware.